HiRes Processing Pipeline

APPN Aerial SOP — Locked revision 1.03

HiRes Processing Pipeline

Caution

Portions of this method are **still under development**, with work continuing over the 2026 winter season. The flight parameters in the HiRes Fieldbook were chosen to allow the different processing methods to be tested robustly.

Important

This page documents the **standardised GRYFN processing pipelines** used within APPN for UAV-based data processing, intended for trained APPN staff processing CALViS, GOBI, HiRes, and M3M datasets. Adherence to the pipelines below ensures reproducibility, quality assurance, and cross-project comparability. For any processing run that **deviates from the standard pipeline**, detailed records must be kept of every parameter changed, the rationale, and any anticipated implications for data quality.

This page documents the standardised HiRES (PhaseOne) processing pipeline within APPN for UAV-based data processing. These configurations define consistent, transparent, and repeatable processing workflows across GRYFN-supported sensors and platforms, supporting reproducibility, quality assurance, and cross-project comparability. The files are intended to be used as reference and default templates for approved processing pipelines, with any deviations explicitly documented to maintain traceability and data integrity.

The standard pipelines and the data products they output are detailed below. For a tabular summary of output formats, resolutions and software compatibility, see Standard Data Products.

Document Structure

- Processing methods
- Method 1 — Orthomosaic in Agisoft Metashape
- Image tiling methods (in development)
- PhaseOne iX Capture (Windows GUI)
- PhaseOne Image SDK (Linux CLI / shell scripts)

Processing methods

APPN's HiRes (PhaseOne) processing approach is still being developed. The method documented in full below is **Method 1 — an orthomosaic workflow in Agisoft Metashape**, which currently serves as a practical, hands-on starting point for working with HiRes data.

There are also two **image tiling methods**, based on the PhaseOne iX Capture / Image SDK toolchain, which were developed independently by **UWA/DPIRD** and **UQ**. Ongoing work over the season is focused on combining them into a single, unified approach, which will be documented in future revisions of this protocol.

#	Method	Status	Developed by	Toolchain	Operating system	Primary output
1	Orthomosaic in Agisoft Metashape	Documented (interim)	APPN	Agisoft Metashape	Windows or Linux	RGB orthomosaic + DEM
—	Image tiling — PhaseOne iX Capture (GUI)	In development	UWA/DPIRD	PhaseOne iX Capture (GUI)	Windows	Calibrated, tiled RGB frames
—	Image tiling — PhaseOne Image SDK (CLI)	In development	UQ	PhaseOne Image SDK + shell scripts	Linux	Calibrated, tiled RGB frames

Note

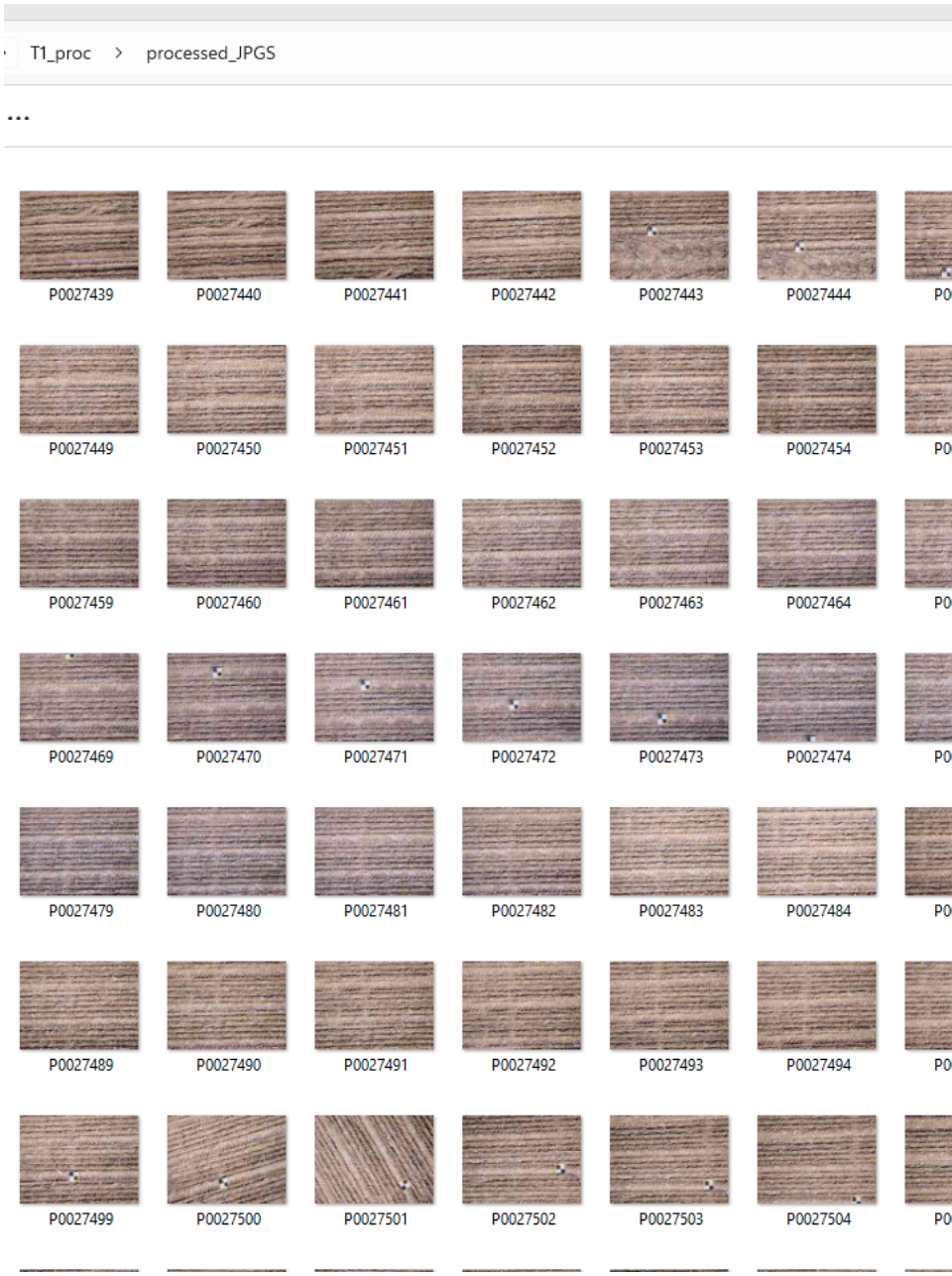
Method 1 is an **interim** workflow for getting familiar with the data; the APPN-uniform approach has not yet been confirmed. Its main advantage is that it is **easy to implement** and produces a **single data product** that is straightforward to work with and to display. The trade-off is that the stitching process introduces some **mosaicing artefacts** and a slight **loss of resolution**.

The two **image tiling methods** were developed independently by UWA/DPIRD and UQ. Both apply the same PhaseOne radiometric calibration, lens correction, and geometric model, and ongoing work this season is focused on reconciling them into a **single, unified approach**.

Method 1 — Orthomosaic in Agisoft Metashape

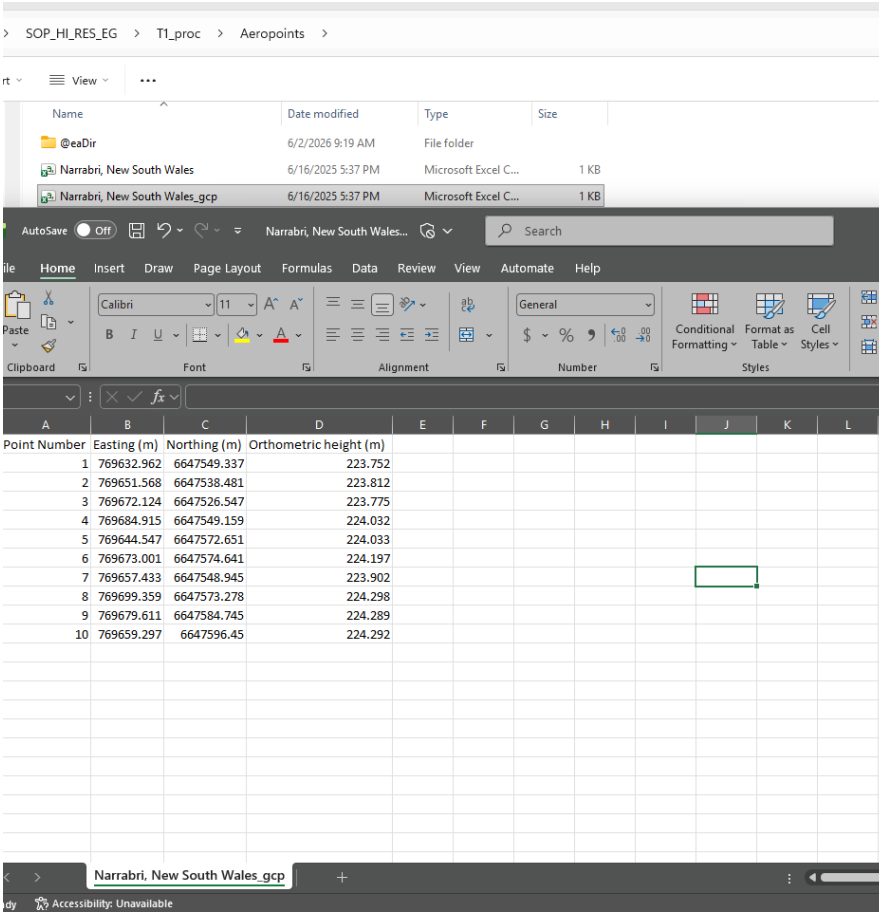
Note

This is an interim, hands-on workflow so those new to HiRes can get a feel for the data and Agisoft. It is not perfect and is not a one-size-fits-all approach. The APPN-uniform approach has not been confirmed as of yet. — Richard Harwood



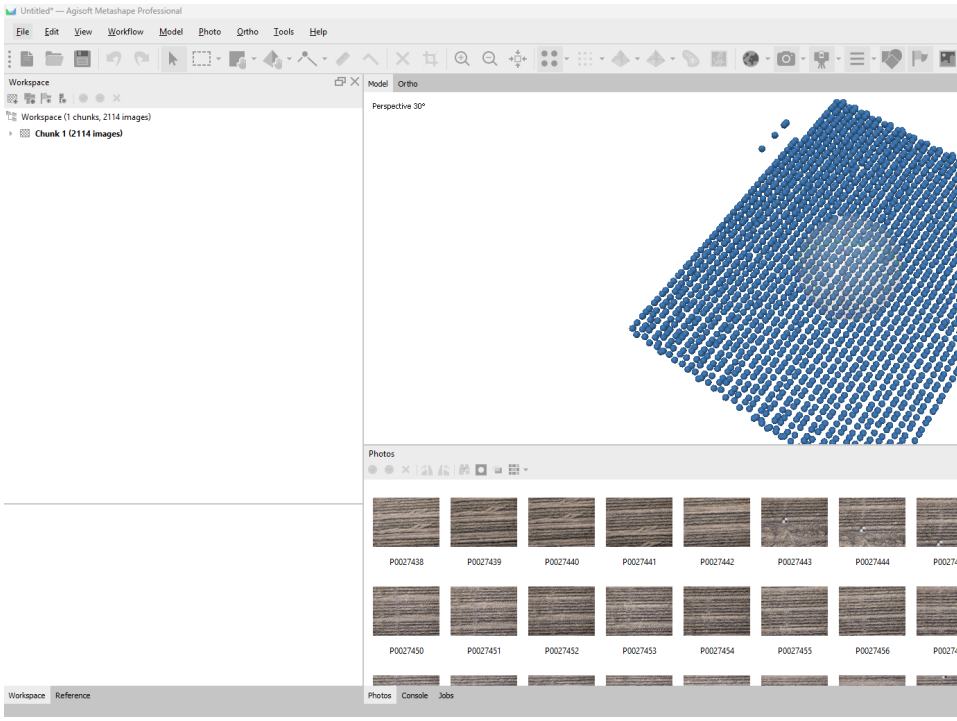
Step 1 — Locate the processed JPEGs

Start with a folder of the processed HiRes JPEG frames ready for import.



Step 2 — Prepare the AeroPoints / GCP file

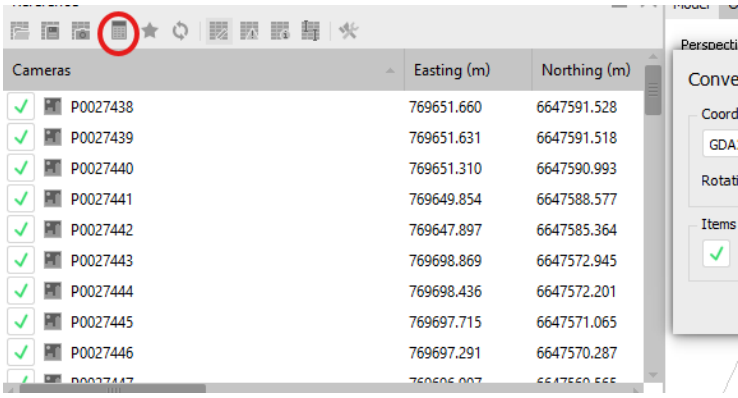
Format the AeroPoints (ground control points) into the layout Metashape expects — label, easting/northing (or lat/lon), and elevation.



Step 3 — Load the JPEGs into Agisoft

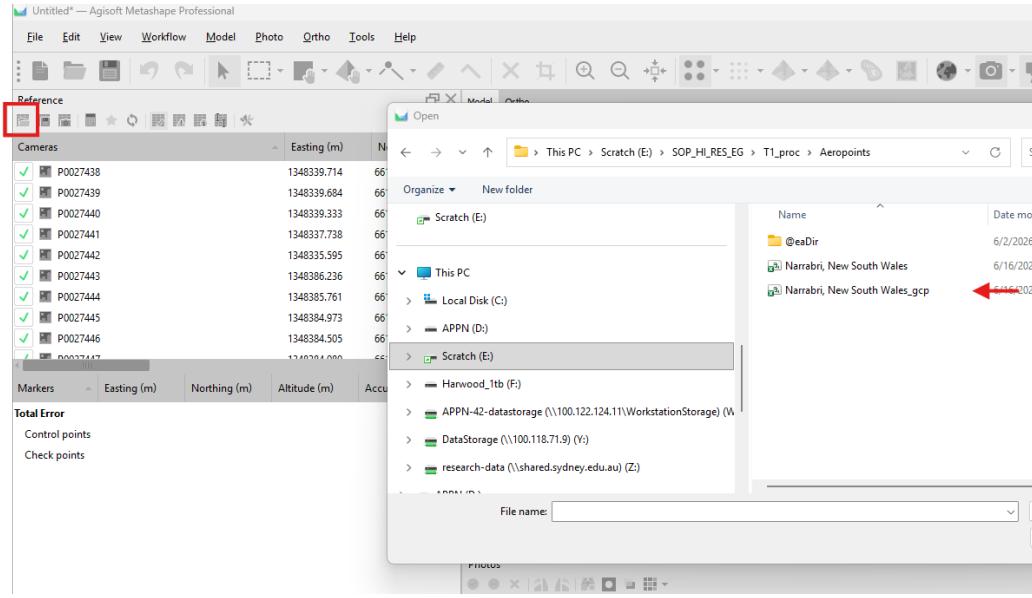
Drag and drop (or **Add Photos**) the processed JPEGs into the Metashape chunk so the images appear in the

workspace.



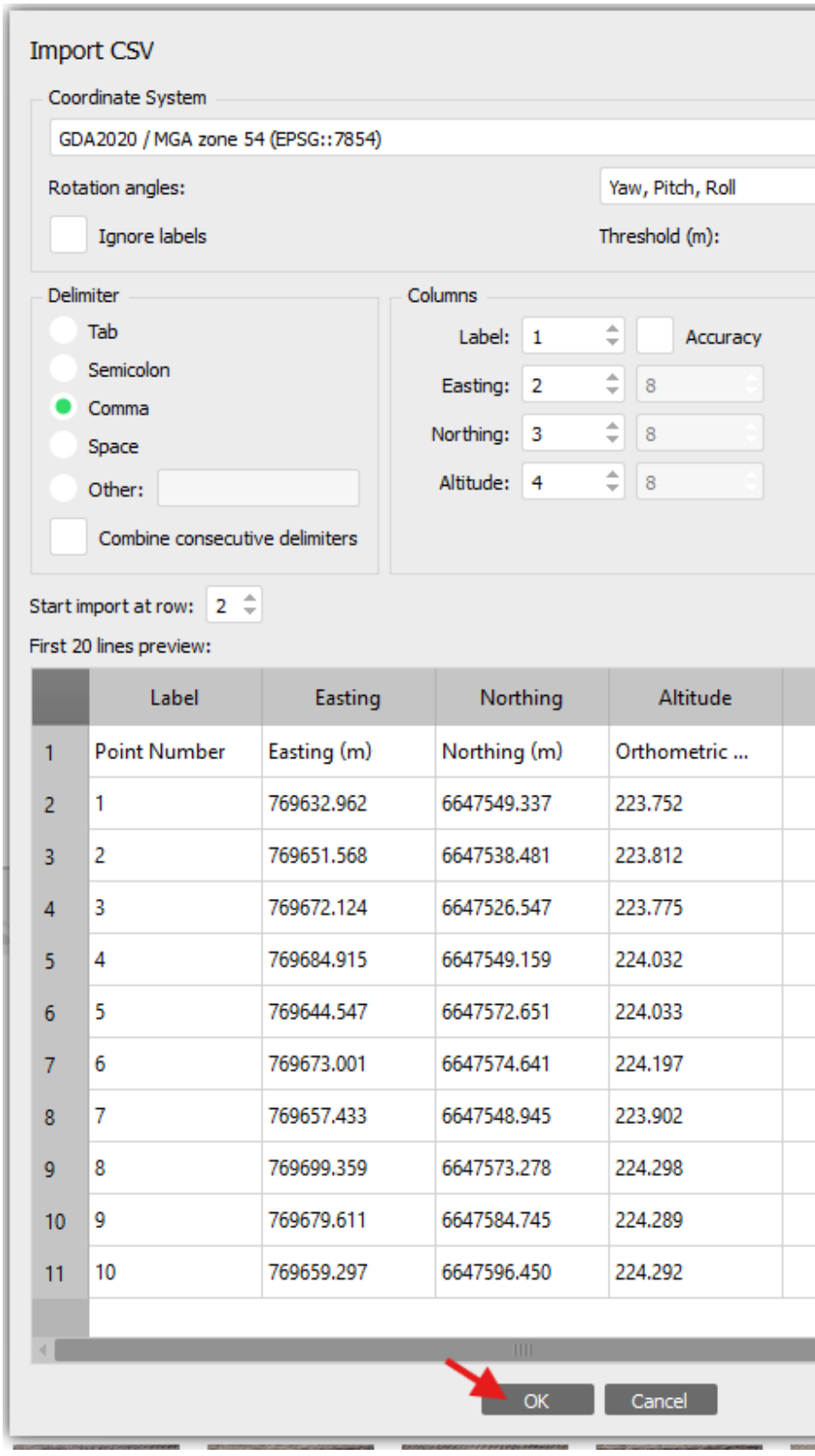
Step 4 — Set / convert the coordinate reference system

Set the camera CRS and convert it to the project's target CRS so the imagery and control points share the same coordinate system.



Step 5 — Import the AeroPoints

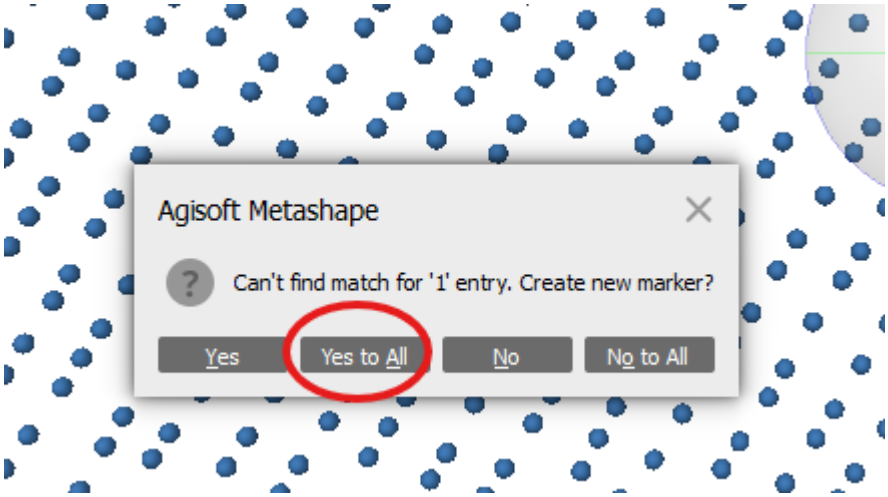
Import the formatted AeroPoints file as markers/GCPs, mapping the columns to the correct coordinate fields.



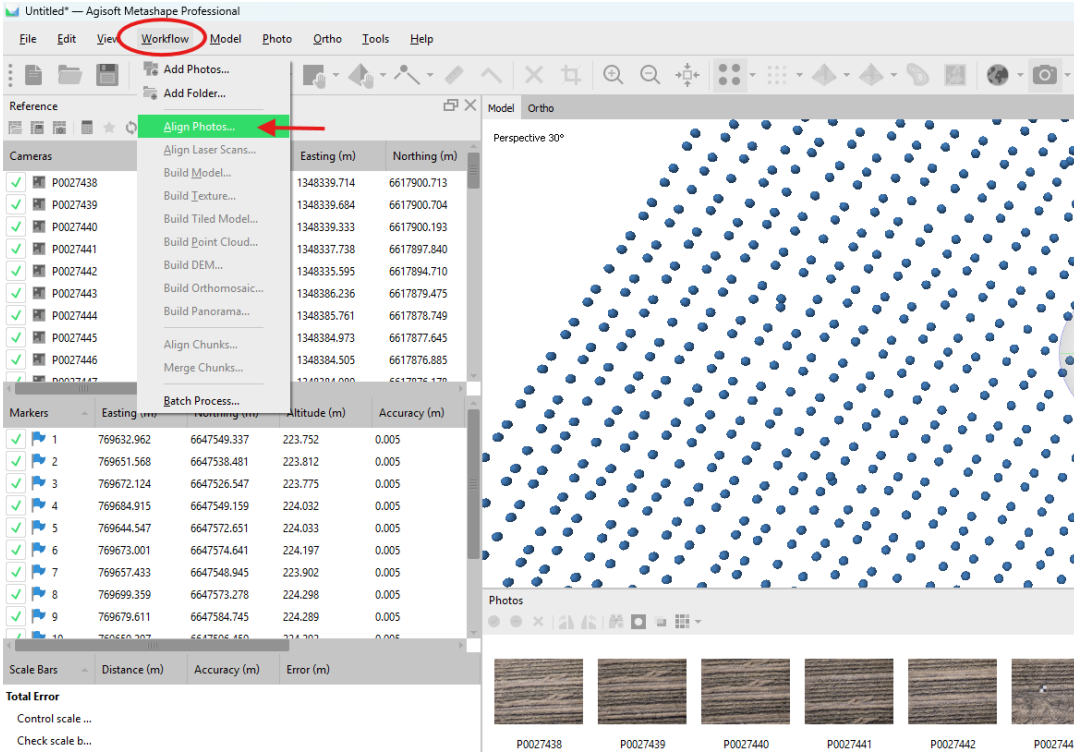
Step 6 — Check the imported AeroPoints

Verify the markers loaded correctly and sit in sensible positions relative to the camera locations.

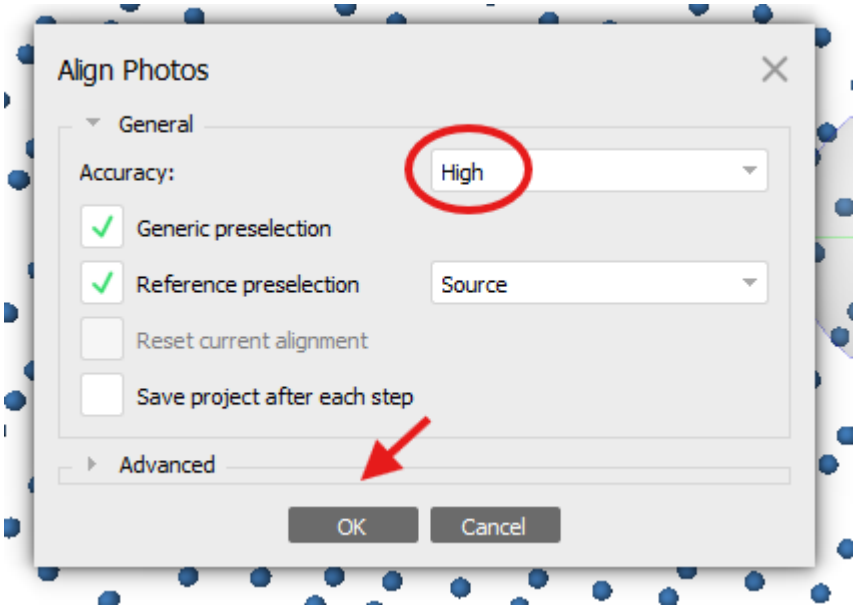
Step 7 — Hit "Yes to all"



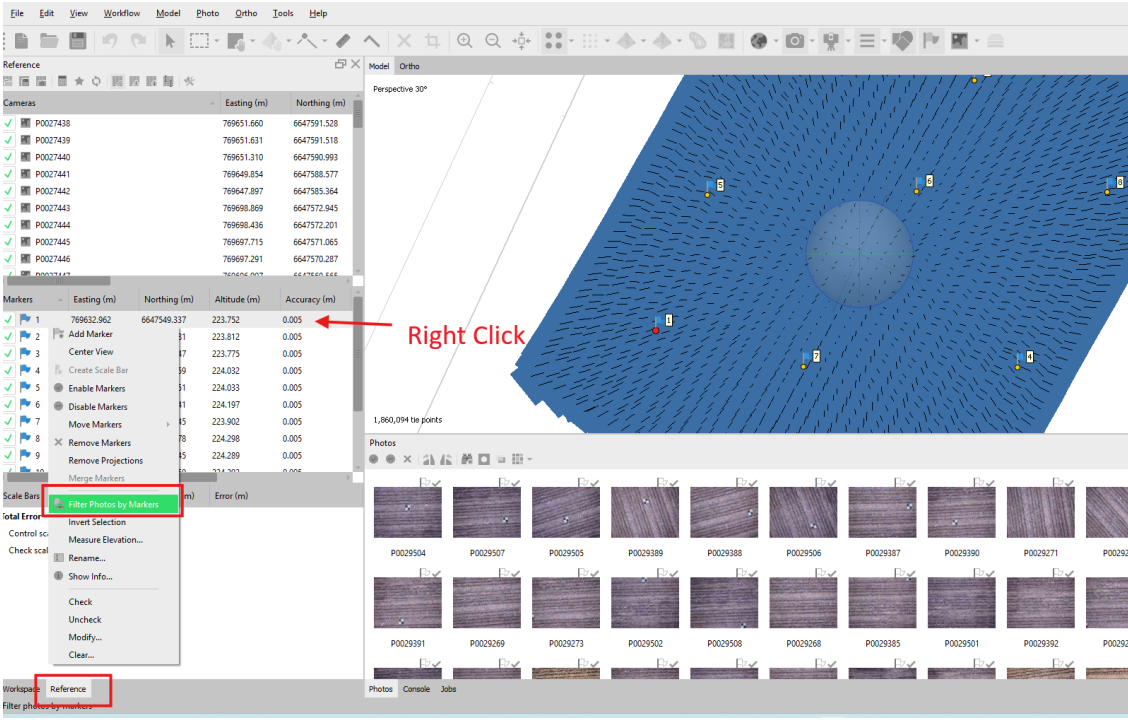
Step 8 — Set up Align photos



Run **Align Photos** to estimate camera positions and build the sparse point cloud.



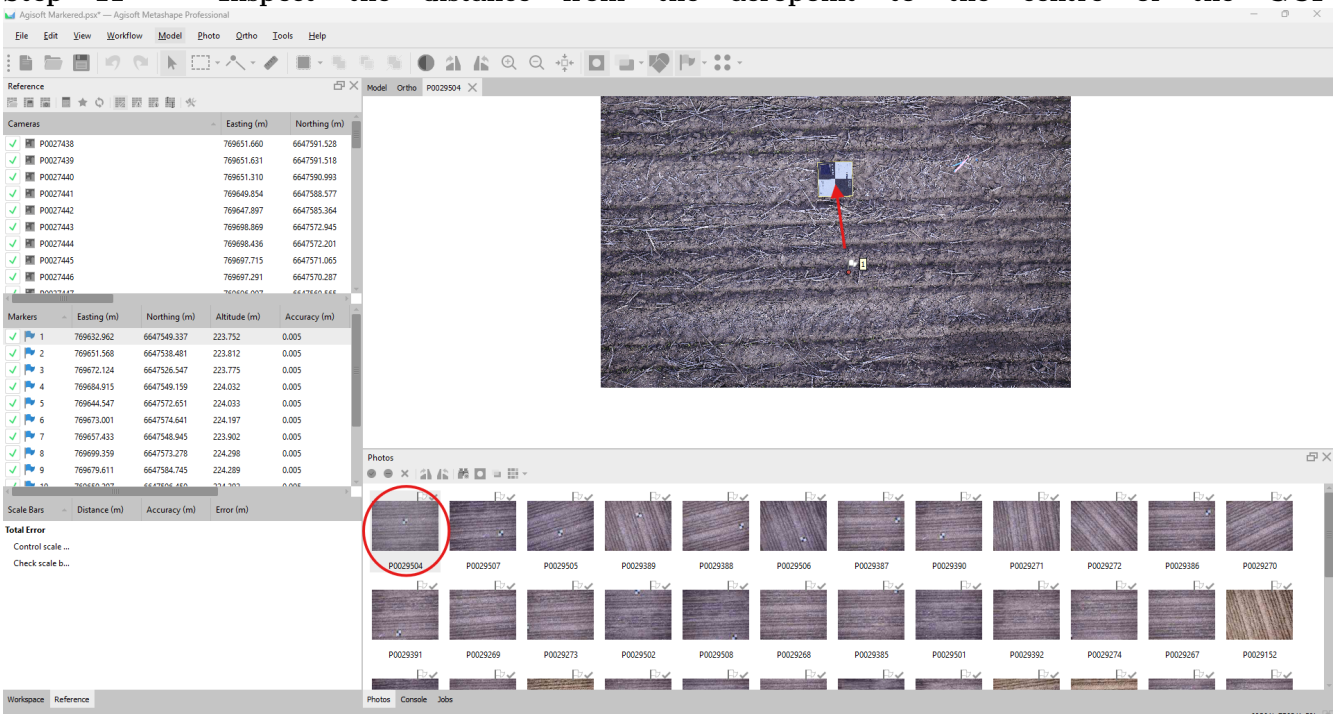
Step 9 — Run Alignment

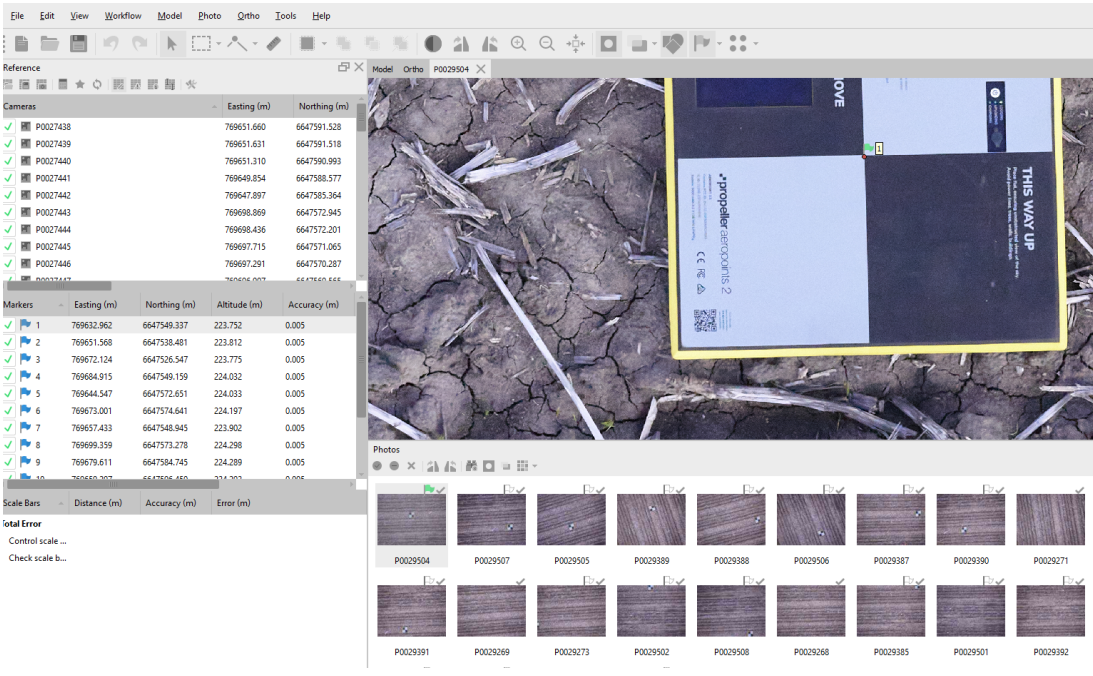


Step 10 — Load a marker

Select a marker (GCP) to begin refining its position across the images. Select filter photos by marker to see which images contain the marker.

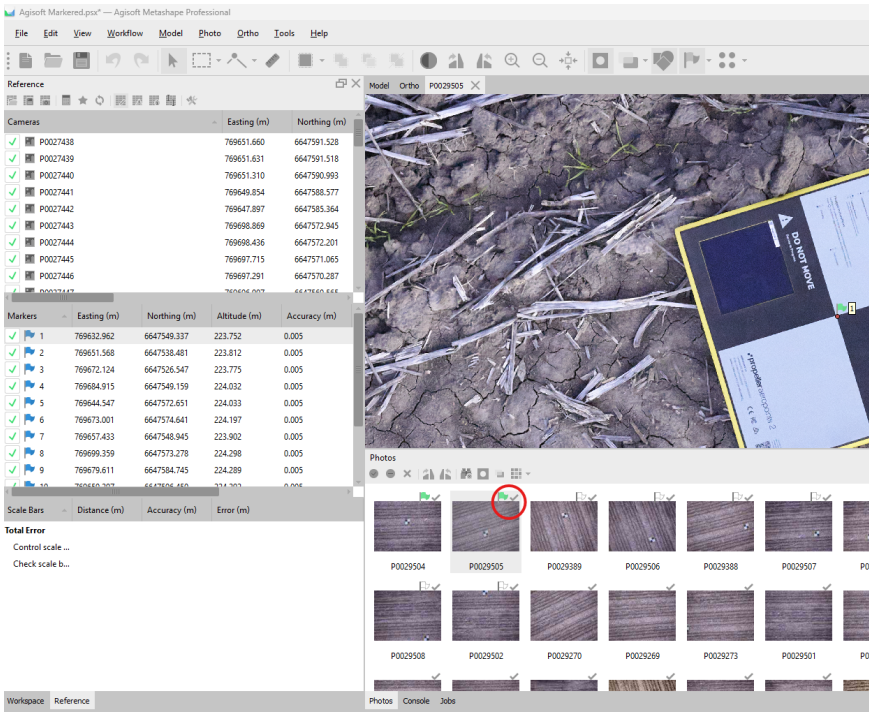
Step 11 — Inspect the distance from the aeropoint to the centre of the GCP





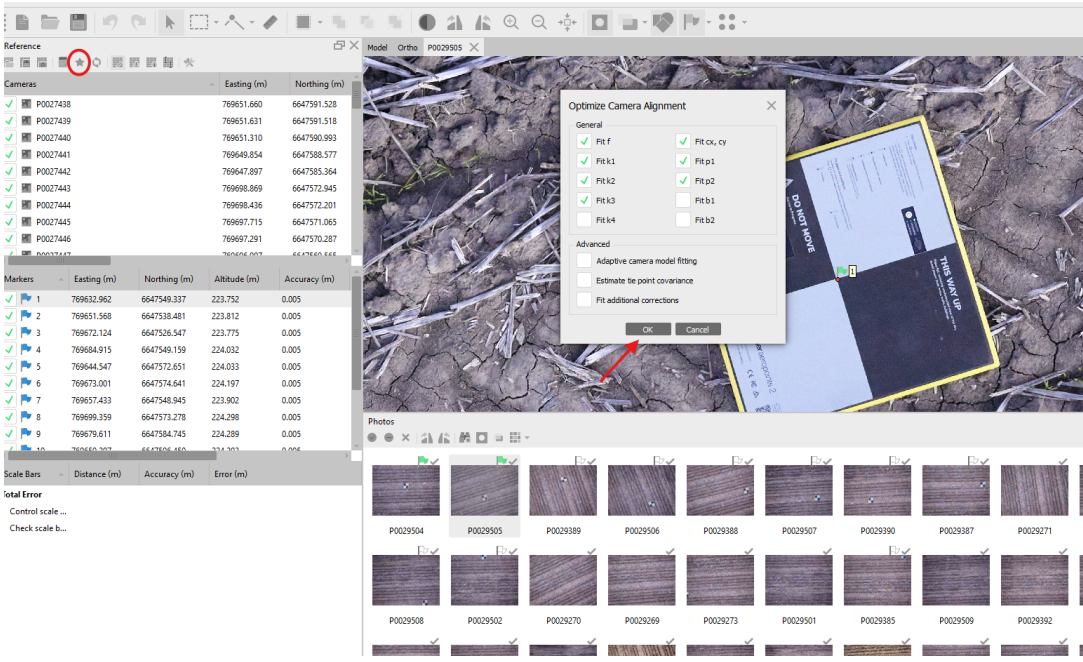
Step 12 — Move the marker

Drag the marker to its precise location on the ground control target in the image.



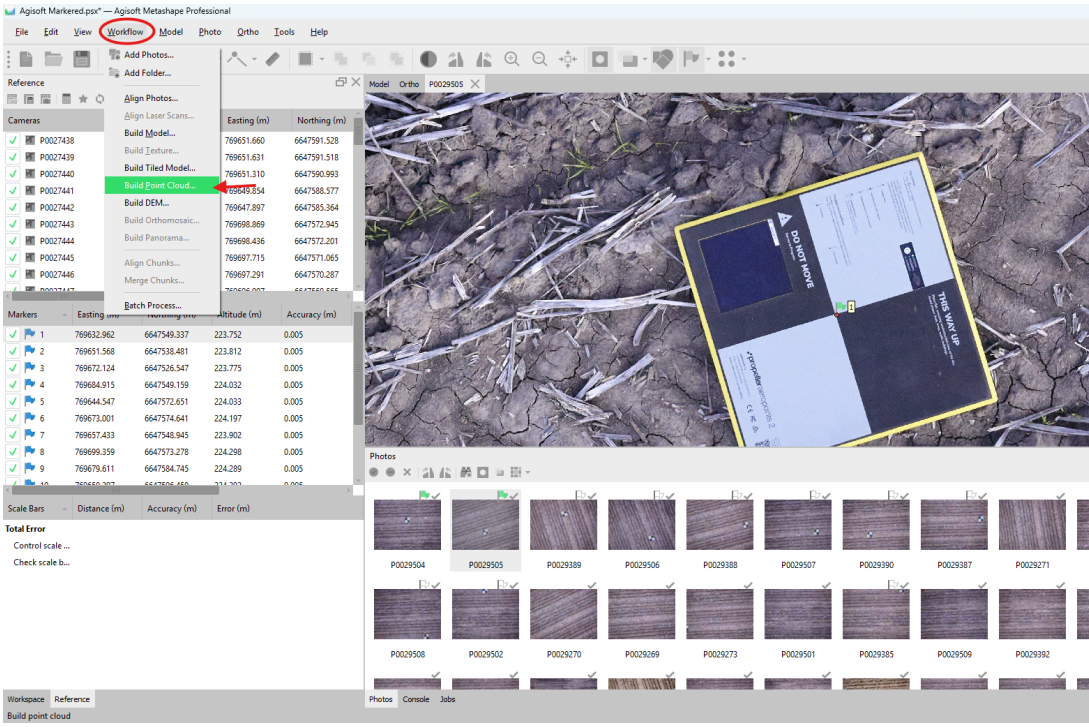
Step 13 — Repeat across images and markers

Repeat the marker placement on the remaining images so each GCP is accurately pinned across all views. Note that you should do ~ 5 images fro a GCP/Marker. Essentially repeat the process until the point is consistently falling in the middle of the GCP for a given marker.



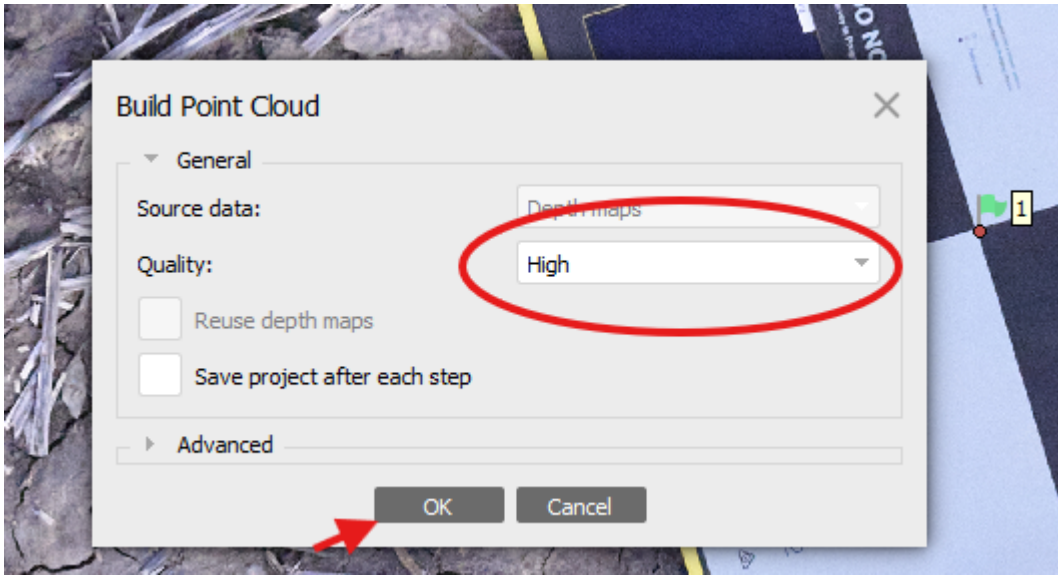
Step 14 — Optimise cameras

Run **Optimise Cameras**



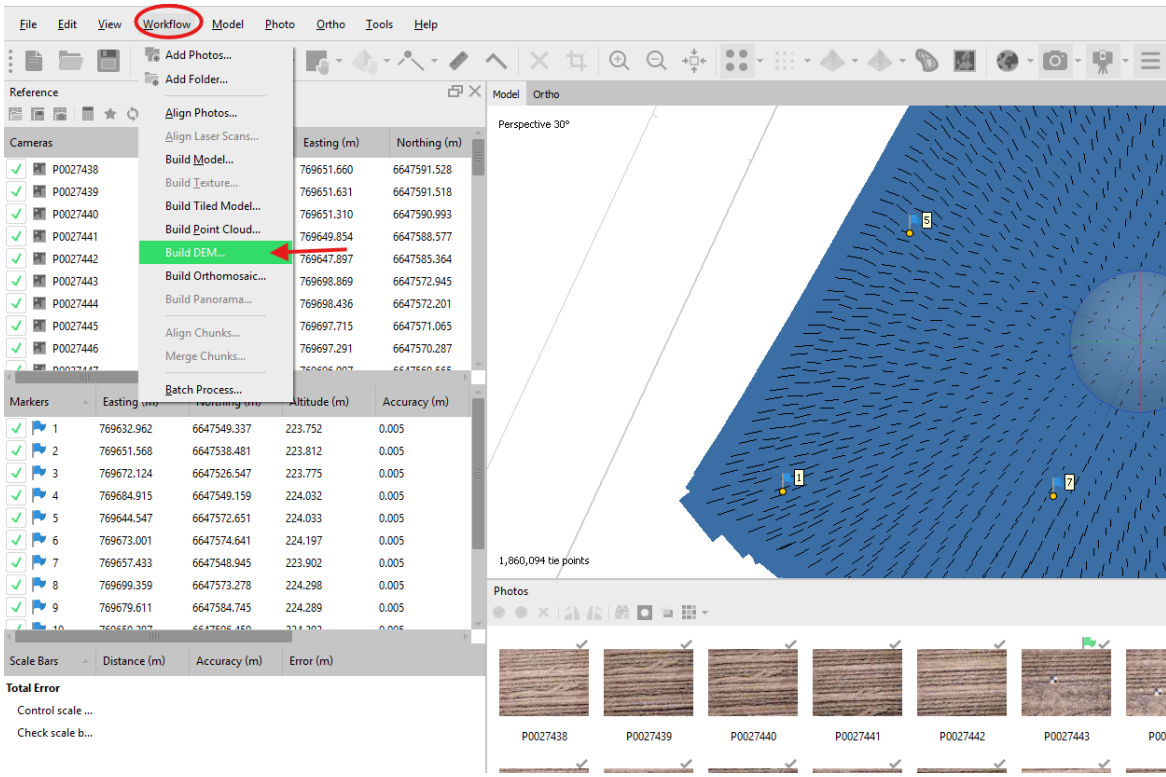
Step 15 — Build point cloud

Start **Build Point Cloud** to generate the dense point cloud from the aligned imagery.



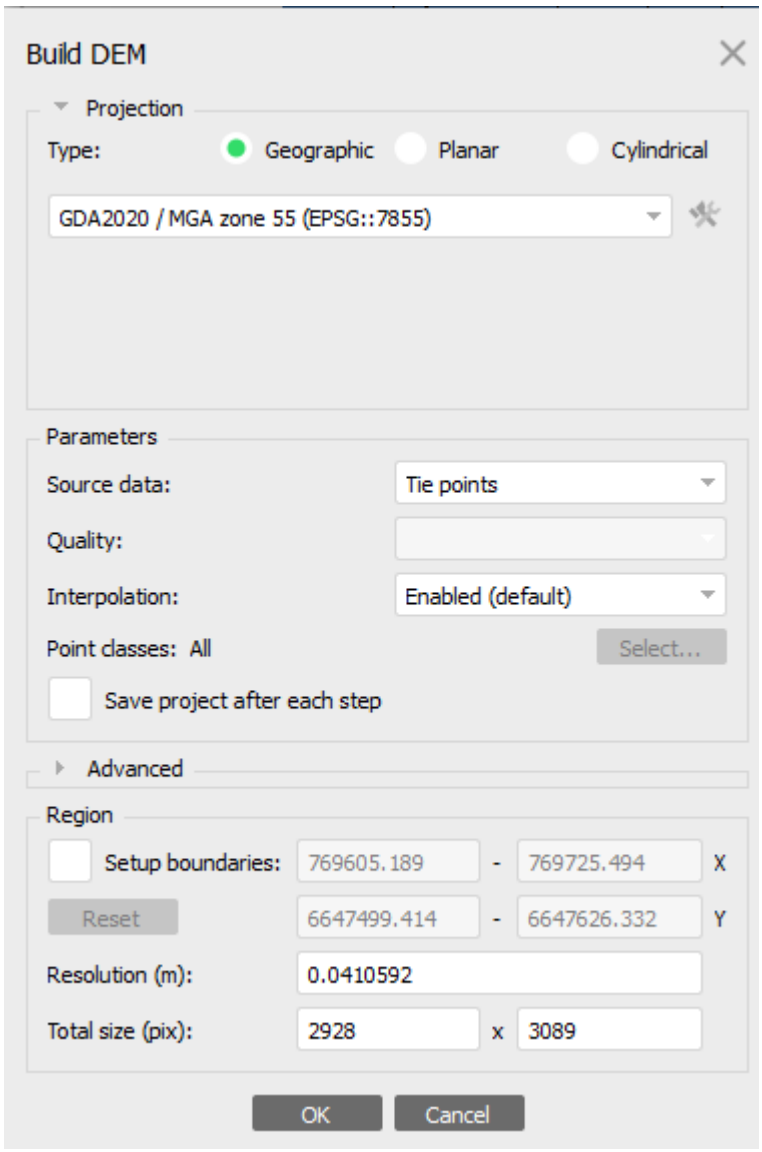
Step 16 — Point cloud settings

Set the quality and depth-filtering options, then run the dense point cloud build.



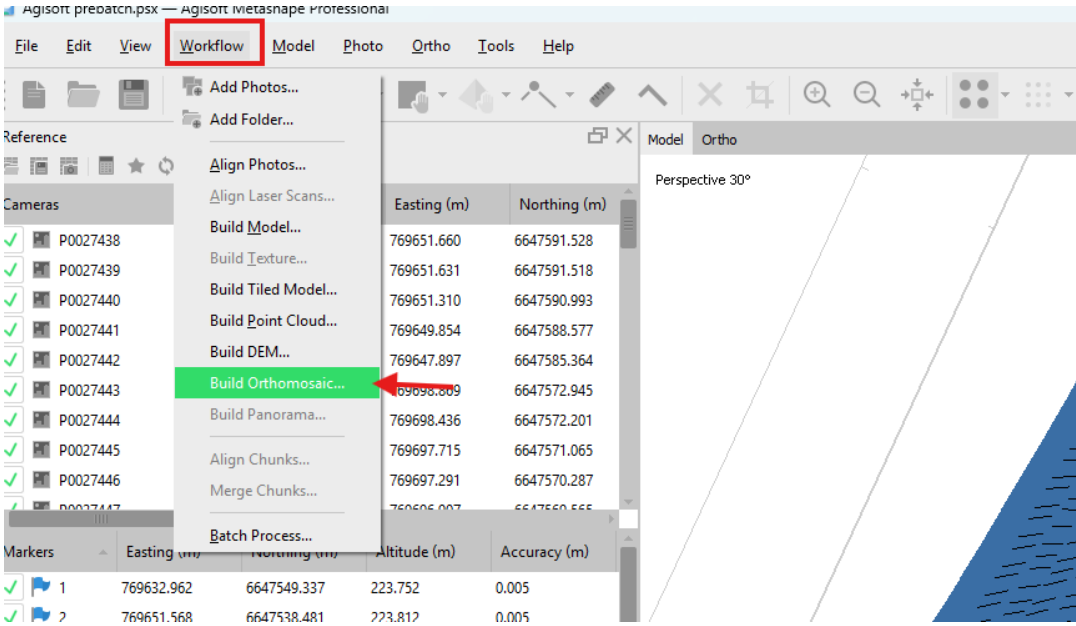
Step 17 — Build DEM

Start **Build DEM** to generate the digital elevation model from the point cloud.



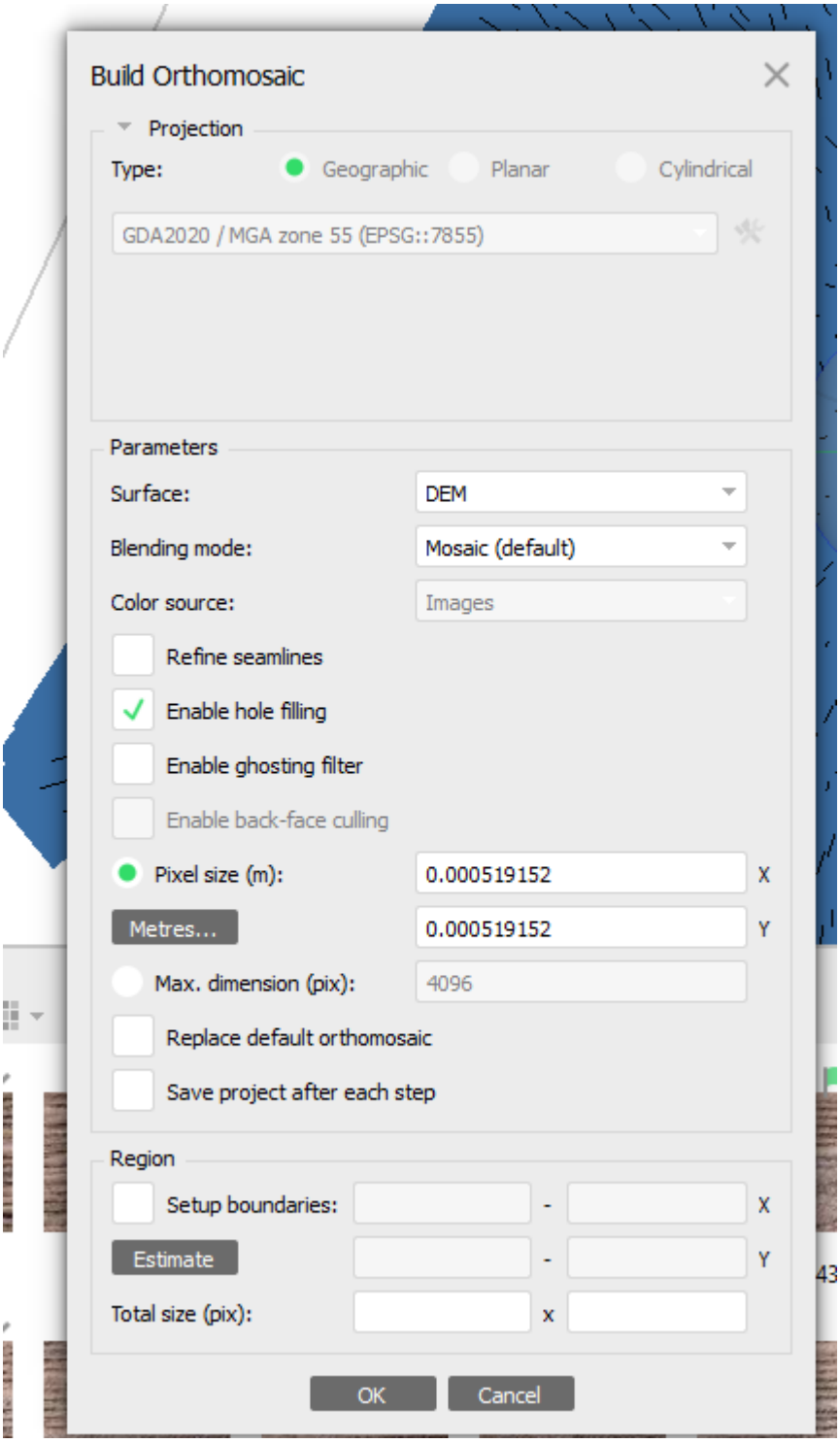
Step 18 — DEM settings

Confirm the source data, projection, and resolution, then run the DEM build.



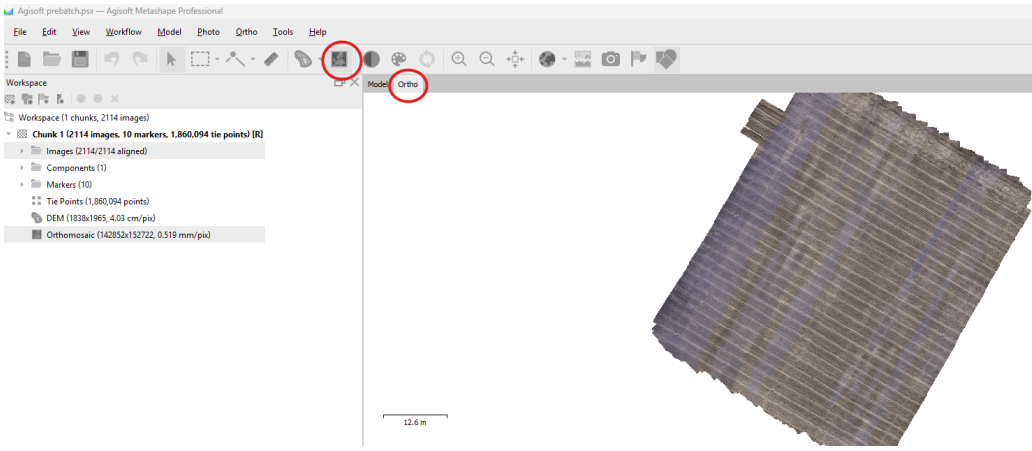
Step 19 — Build orthomosaic

Start **Build Orthomosaic** to generate the orthorectified mosaic from the imagery and DEM.



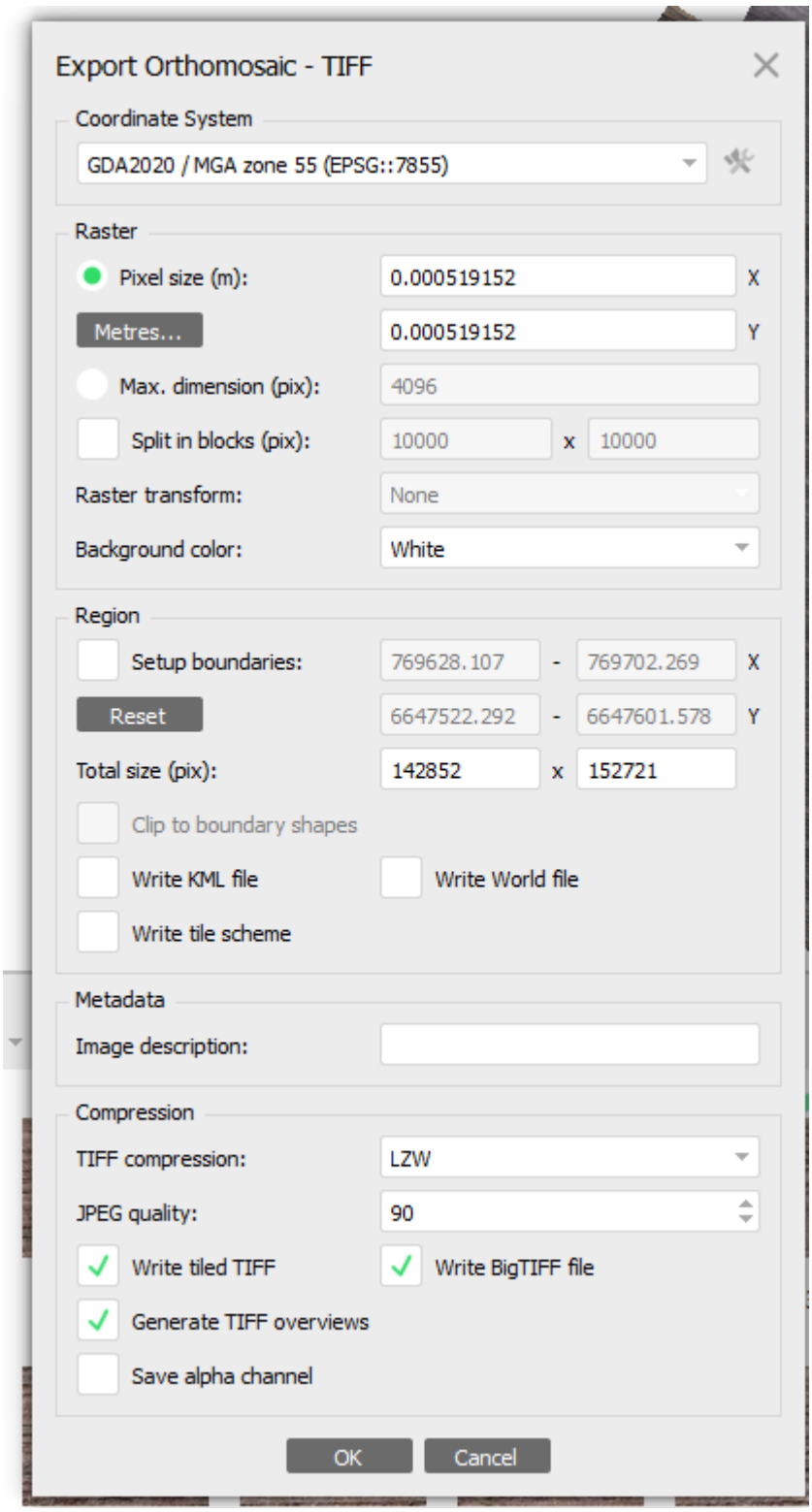
Step 20 — Orthomosaic settings

Set the surface (DEM), blending mode, and resolution, then run the orthomosaic build.



Step 21 — View the orthomosaic

Inspect the finished orthomosaic for coverage, colour balance, and any mosaicing artefacts.



Step 22 — Export the orthomosaic

Export the orthomosaic as a GeoTIFF, confirming the CRS, resolution, and output path.

Method 2 - Image tiling method (in development)

> **Important**
>
> The two methods have were developed **independently by UWA/DPRID and UQ** and are based on the PhaseOne iX Capture / Image SDK toolchain. Ongoing work this season is focused on **combining them into a single, unified approach**, which will be documented in future revisions of this protocol. Both are intended to produce the standard, radiometrically and geometrically rigorous APPN tiled image products.